

Multiple Choice

1. **Answer:** E

Options: A) $\text{AlCl}_3(\text{aq})$; B) $\text{CBr}_4(\text{aq})$; C) $\text{NaOH}(\text{aq})$; D) $\text{HCl}(\text{aq})$; E) $\text{LiOH}(\text{aq})$

Note: Correct option: E - $\text{LiOH}(\text{aq})$

2. **Answer:** A

Options: A) 4.26; B) 3.66; C) 9.26; D) 2.66; E) 10.52

Note: Correct option: A - 4.26

3. **Answer:** A

Options: A) $400 \text{ N} \cdot \text{m}^{-1}$; B) $350 \text{ N} \cdot \text{m}^{-1}$; C) $250 \text{ N} \cdot \text{m}^{-1}$; D) $285 \text{ N} \cdot \text{m}^{-1}$; E) $450 \text{ N} \cdot \text{m}^{-1}$

Note: Correct option: A - $400 \text{ N} \cdot \text{m}^{-1}$

4. **Answer:** A

Options: A) 1.75; B) 3.50; C) 14.00; D) 10.00; E) 12.00

Note: Correct option: A - 1.75

5. **Answer:** A

Options: A) $3.0 \times 10^5 \text{ Joules}$, $3.0 \times 10^{12} \text{ ergs}$; B) $2.5 \times 10^5 \text{ Joules}$, $2.5 \times 10^{12} \text{ ergs}$; C) $2.0 \times 10^5 \text{ Joules}$, $2.0 \times 10^{12} \text{ ergs}$; D) $1.8 \times 10^5 \text{ Joules}$, $1.8 \times 10^{12} \text{ ergs}$; E) $3.56 \times 10^5 \text{ Joules}$, $3.56 \times 10^{12} \text{ ergs}$

Note: Correct option: A - $3.0 \times 10^5 \text{ Joules}$, $3.0 \times 10^{12} \text{ ergs}$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'at least four sigma bonds'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $a' = 2 RT_c V_c$, $b' = (V_c / 2)$ '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 5. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 86.2 *Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.*
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key